Use-Case Flow Specification

Use Case: UC-04 — Allocate ER/ICU Bed

Primary Actor: Triage Nurse

Supporting Actor: Hospital Administrator

1. Preconditions

1. Patient intake has been recorded and triage severity has been computed.
2. The dashboard is connected to the current ER/ICU bed-status data.
3. The Triage Nurse is authenticated and has permission to allocate a bed.

2. Postconditions

1. A suitable ER/ICU bed is assigned to the patient and its status is updated.
2. The patient is placed in the appropriate allocation queue or assigned to an available bed.
3. The updated bed status is visible on connected dashboard clients.

3. Main Success Scenario

1. The Triage Nurse selects the patient requiring bed allocation.
2. The system retrieves the patient's triage severity and current bed requirements.
3. The system checks real-time ER/ICU bed availability.
4. The system identifies suitable available beds and prioritizes the patient according to triage severity.
5. The Triage Nurse selects or confirms the recommended suitable bed.
6. The system assigns the bed to the patient.
7. The system changes the bed status to **Occupied** and updates the dashboard.
8. The system confirms the successful allocation to the Triage Nurse.

4. Alternate Flow

A1 — No suitable bed available: At Step 3, if no suitable ER/ICU bed is available, the system keeps the patient in the prioritized allocation waitlist and notifies the appropriate staff. The administrator may initiate an inter-facility ambulance transfer when required.

A2 — Bed becomes unavailable: If the selected bed becomes unavailable before confirmation, the system rejects the allocation and refreshes the real-time bed list. The Triage Nurse selects another suitable bed.

5. Exception / System Response

If real-time bed-status data cannot be retrieved, the system shall not make an allocation using stale information. It shall notify the user and request a refreshed connection/status before proceeding.